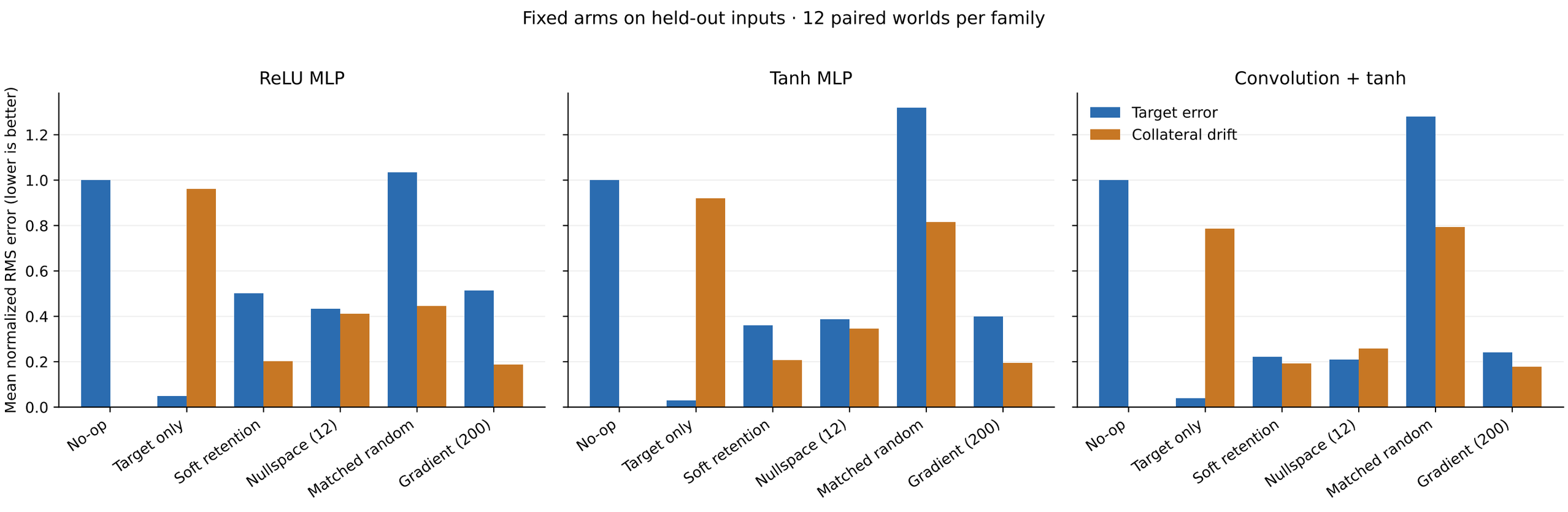
Fixed arms on held-out inputs · 12 paired worlds per family
ReLU MLP
Tanh MLP
Convolution + tanh
Mean normalized RMS error (lower is better)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Target error
Collateral drift
No-op
Target only
Soft retention
Nullspace (12)
Matched random
Gradient (200)